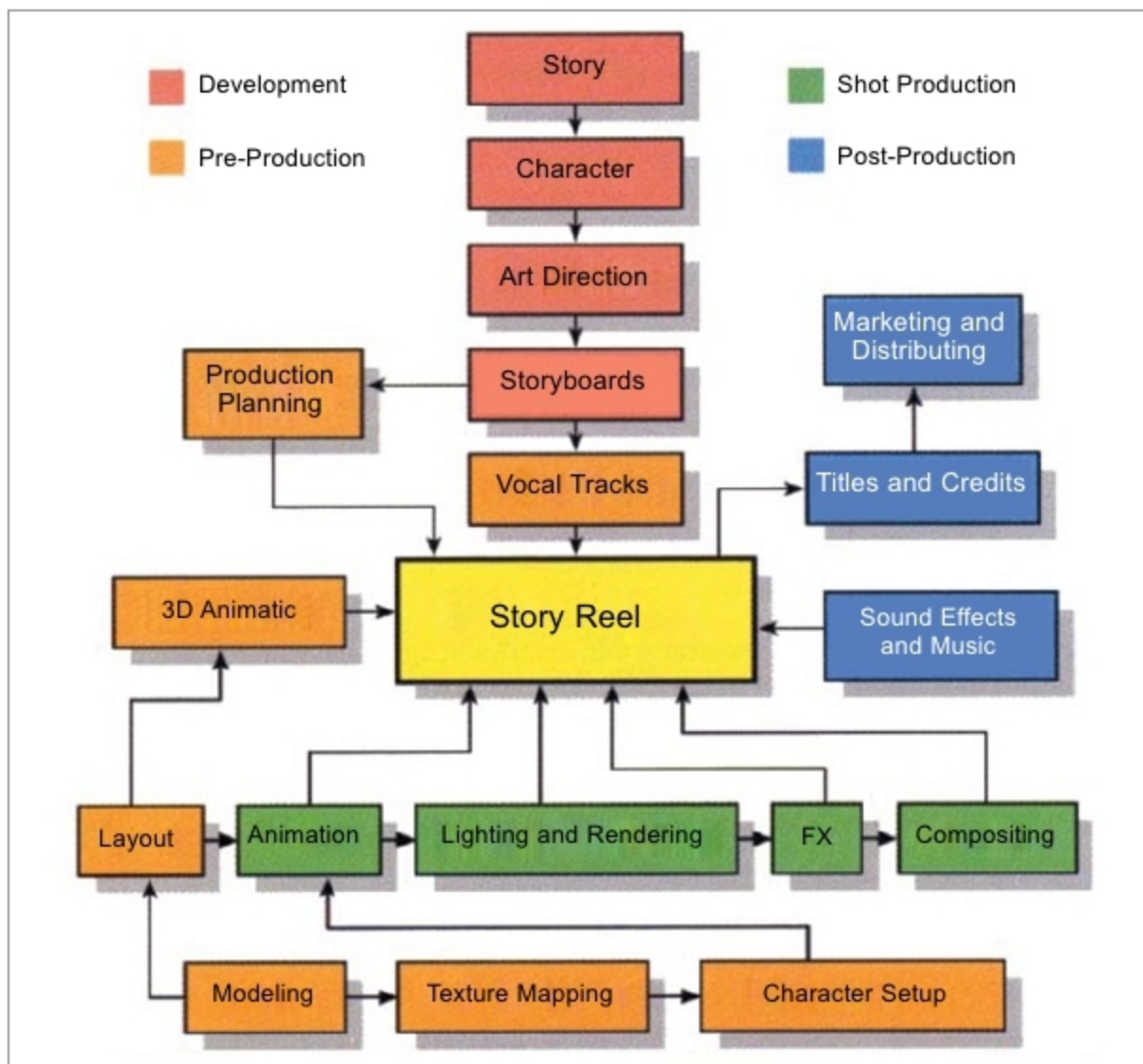




Steps for creating a 3D animation (Credit: Scanta)

tasks are completed at predetermined stages before moving to the next step. It is like an assembly line for animated film production. The process is broken down into three major phases: pre-production, production and post-production. These phases incorporate tasks that need to be completed before moving on to the next one. Each step builds on previous tasks that were completed.

Phases of 3D animation

Steps involved in creating a 3D animation are explained below.

Pre-production. The first step in the process of creating a 3D animation is conceptualisation of an idea and translation of said idea into visual form. Pre-production has two dimensions; the first is deciding what will happen first, next and last.

The second is interaction, which involves deciding how the voiceover will interact with images, how visual transitions and effects will help tie images together and how voiceovers will interact with the musical soundtrack.

3D modelling. 3D modelling is a technique in computer graphics for

producing a 3D digital representation of any object or surface. A 3D object can be generated automatically or created manually by deforming the mesh or manipulating vertices.

A typical way of creating a 3D model is to take a simple object, called primitive, and extend or grow it into a shape that can be refined and detailed. A primitive can be anything from a single point (called vertex), a two-dimensional line (edge), a curve (spline) or a 3D object (face or polygon).

Texturing. The next step is shading and texturing. In this phase, textures, materials and colours are added to the 3D model. Every component of the model receives a different shader material to give it a specific look.

Realistic materials. If the object is made of plastic, it is given a reflective, glossy shader. If it is made of glass, the material is partially transparent and refracts light like real-world glass.

Rigging. Rigging is what makes deforming a character possible. It includes creating a skeleton to deform the character and creating animation controls to enable easy posing of the character. Rigging usually involves adding bones, calculating and imple-

menting skin weights, and adding muscles to create natural movements.

Animation. This is a process of taking a 3D object and making it move. It is up to the animator to make a 3D object feel like it is alive and breathing.

Rigged for motion. A 3D character is controlled using a virtual skeleton or rigging, which allows the animator to control the model's arms, legs, facial expressions and postures.

Pose-to-pose. Animation is typically completed pose-by-pose.

Baking. Before an animation can be used in Unity software, it must first be imported to the project. Unity can import native Maya (.mb or .ma) and generic FBX files, which can be exported from most animation packages. FBX files include models, bones and animations made in 3D software. There are different texture files (jpeg/png) that are included at the time of exporting project from Maya to Unity.

Unity software

Unity is a cross-platform game engine. Steps involved in creating a 3D animation using Unity are given below.

Importing FBX files and textures in Unity. Assets are imported to Unity Editor and all their textures are extracted. Importing these files comes with two things, namely, materials and textures. Materials refer to plastic or glass.

Making prefab of 3D model. Unity allows users to intuitively create a duplicate, called prefab. Prefab allows storing an asset with all its properties inside the prefab. It acts as a template that designers can use to create new instances of the same object in the scene.

Customising prefab according to requirement. This involves:

- Adding events according to frame rate
- Adding audio to animation
- Adding shading and lights to animation

Adding assets to folder and naming it. An asset is a representation of any item a designer can use in a game or project. It may come from a file created outside Unity, such as a 3D model, audio file, image or any other file type that Unity supports.

Every communication device